

Car Price Prediction Model using Python

Aim

To implement a **Car Price Prediction Model** using Python and **Linear Regression** to predict the selling price of a car based on its features.

Apparatus Required

- Google Colab / Jupyter Notebook
- Python 3
- Libraries:
 - NumPy
 - Pandas
 - Scikit-learn (sklearn)

Theory

Car Price Prediction is a **supervised machine learning regression** problem. The objective is to predict the selling price of a car based on features such as the manufacturing year, kilometers driven, fuel type, and engine power.

In this experiment, **Linear Regression** is used to learn the relationship between the input features and the car price.

Advantages:

- Simple and easy to implement.
- Fast training and prediction.
- Suitable for regression problems.

Disadvantages:

- Assumes a linear relationship.
- Performance decreases on highly non-linear data.

Algorithm

1. Import the required libraries.
2. Create a sample car dataset.
3. Split the dataset into training and testing sets.
4. Train the Linear Regression model.
5. Predict the selling prices.
6. Calculate the Mean Squared Error (MSE) and R^2 Score.
7. Display the predicted prices.

Code (Google Colab)

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample Car Dataset
data = {
    'Year': [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022],
    'Kilometers_Driven': [50000, 40000, 35000, 30000, 25000, 20000, 15000, 10000],
    'Engine_CC': [1200, 1200, 1500, 1500, 1800, 1800, 2000, 2000],
    'Price': [400000, 450000, 550000, 600000, 700000, 800000, 900000, 1000000]
}

df = pd.DataFrame(data)

X = df[['Year', 'Kilometers_Driven', 'Engine_CC']]
y = df['Price']
```

```
# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)
```

```
# Train model
model = LinearRegression()
model.fit(X_train, y_train)
```

```
# Prediction
y_pred = model.predict(X_test)
```

```
# Results
print("Actual Prices:")
print(y_test.values)
```

```
print("\nPredicted Prices:")
print(y_pred)
```

```
print("\nMean Squared Error:")
print(mean_squared_error(y_test, y_pred))
```

```
print("\nR2 Score:")
print(r2_score(y_test, y_pred))
```

Sample Output

```
*** Actual Prices:
[450000 800000]

Predicted Prices:
[428019.32367158 804830.91787446]

Mean Squared Error:
253243949.6822623

R2 Score:
0.9917308098062935
```

Result

The **Car Price Prediction Model** was successfully implemented using **Linear Regression** in Python. The model predicted car prices based on vehicle features, and its performance was evaluated using the **Mean Squared Error (MSE)** and **R² Score**.